



DESIGN AND IMPLEMENTATION OF A

Digital Staff Promotion Support System for GCTU

Secure evidence. Controlled workflow.
Traceable decisions.

PRESENTED BY

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4231230141 | August 2026

FINAL YEAR PROJECT DEFENCE



GCTU

Digital Staff Promotion Support System

Official staff promotion platform



STAFF ACCESS

Sign in

Use your official GCTU account to continue.

Email or username

name@live.gctu.edu.gh

Password

Enter password

Show

Sign In

New lecturer? [Create account](#)

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Official Digital Staff Promotion Support System

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The problem is a broken information flow

The manual process creates delay because evidence, responsibility, and status are not visible in one controlled record.

01

Fragmented evidence

Physical and scattered files are difficult to trace.

02

Slow routing

Movement between offices depends on manual follow-up.

03

Limited visibility

Applicants cannot reliably see where action is required.

04

Weak auditability

Reconstructing who acted, what changed, and when is hard.

RESEARCH PROBLEM

How can GCTU standardise promotion evidence and routing while improving security, transparency, and institutional accountability?

Aim and six objectives

AIM

Design and implement a GCTU-adapted digital platform that supports the academic staff promotion workflow from submission to final recording.

01

Examine bottlenecks

Identify delay, inconsistency, and security risks.

02

Design the workflow

Standardise submission, review, verification, and tracking.

03

Centralise records

Store applications, evidence, feedback, and history securely.

04

Enforce role access

Separate lecturer, academic, HR, committee, and admin duties.

05

Support eligibility

Apply configured rules only to HR-verified evidence.

06

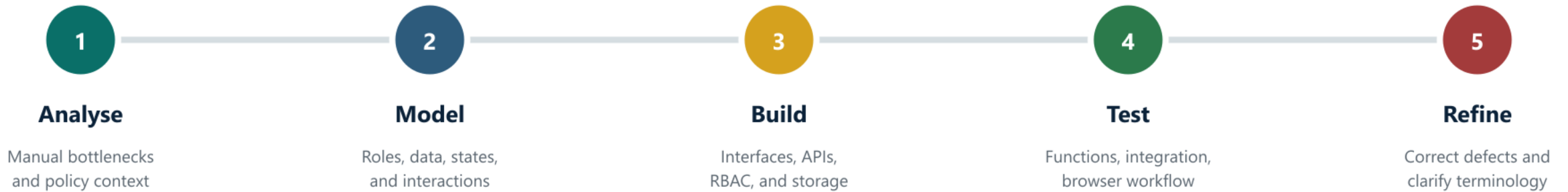
Improve accountability

Provide status, notifications, reports, and audit logs.

RESULT: all six objectives achieved; Objective 5 is explicitly bounded to prototype completeness rules.

Prototyping turned policy into tested workflow

Repeated design, implementation, and evaluation were appropriate because institutional workflow requirements became clearer through a working prototype.



REQUIREMENT SOURCES

Study findings

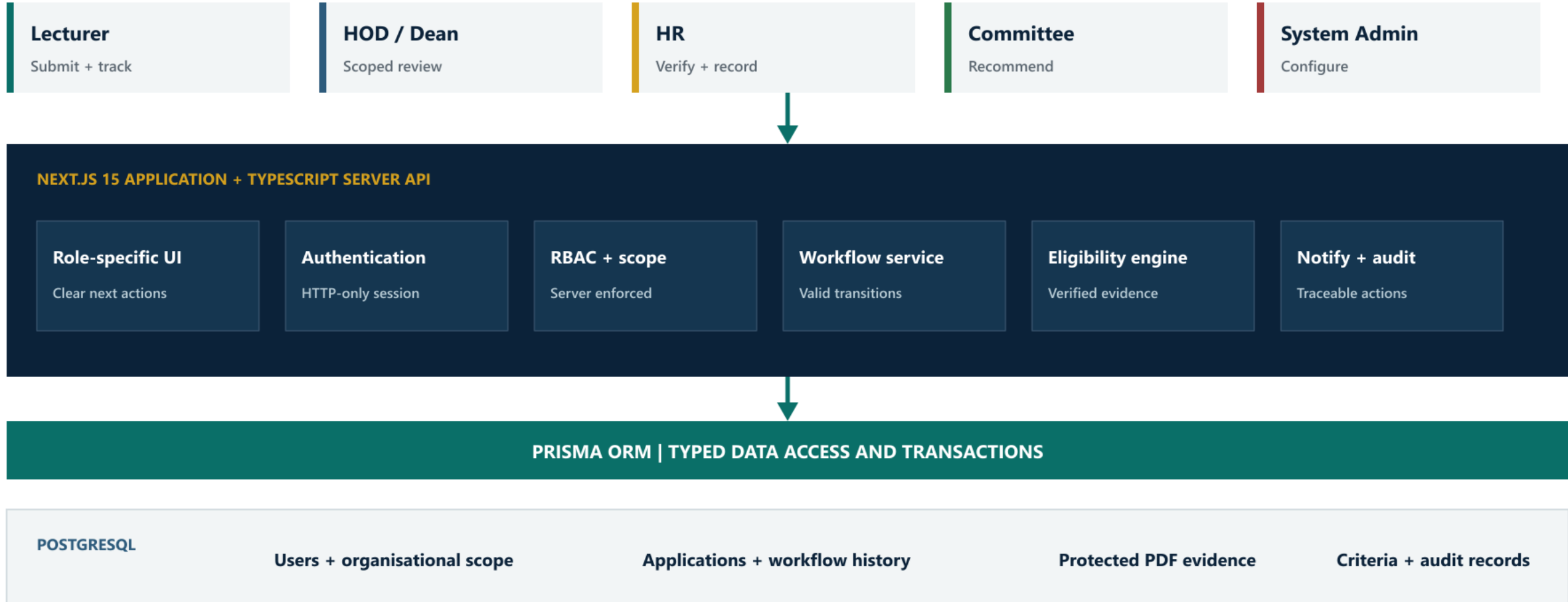
GCTU policy documents

Role responsibilities

End-to-end testing evidence

Layered architecture with server-enforced controls

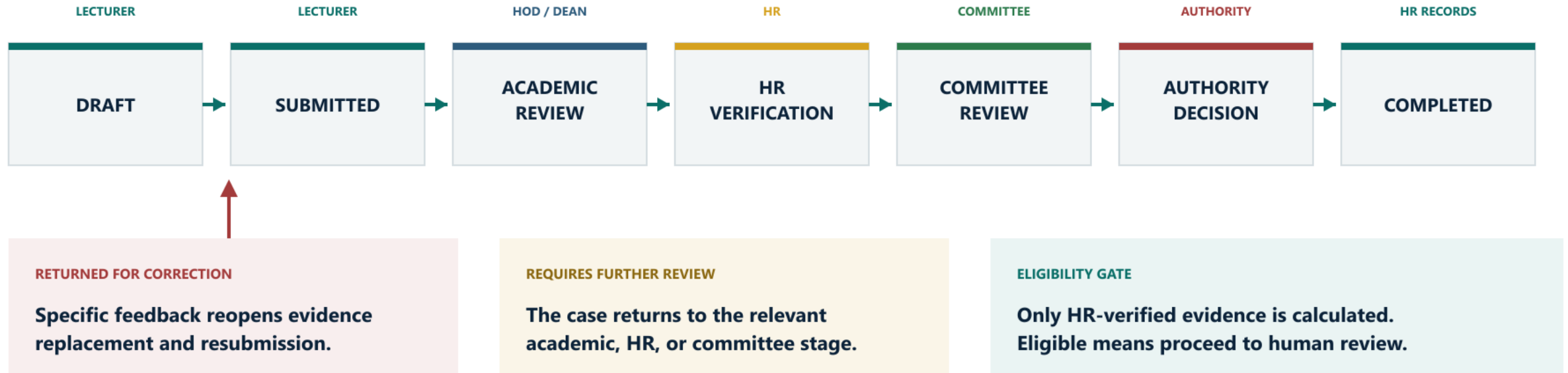
USERS AND ROLE WORKSPACES



Passwords are hashed; access, ownership, scope, validation, and status transitions are checked on the server.

A controlled workflow, not an automatic decision

PRIMARY PROMOTION PATH



ORGANISATIONAL GOVERNANCE

HOD account = department scope

Dean account = faculty scope

Committee recommendation is not final approval

Prototype boundary: HOD and Dean currently share one permission set and one academic-review stage.

Eligibility is evidence completeness, not academic grading

CORE COMPLETENESS SCORE



ALL REQUIRED EVIDENCE GATES

Teaching | Research | Service
Qualifications | Publications | Professional Development

Eligibility permits onward review; it never grants promotion.

THE ENGINE ANSWERS

**Are the configured years-in-rank
and required verified records present?**

HUMAN REVIEWERS ANSWER

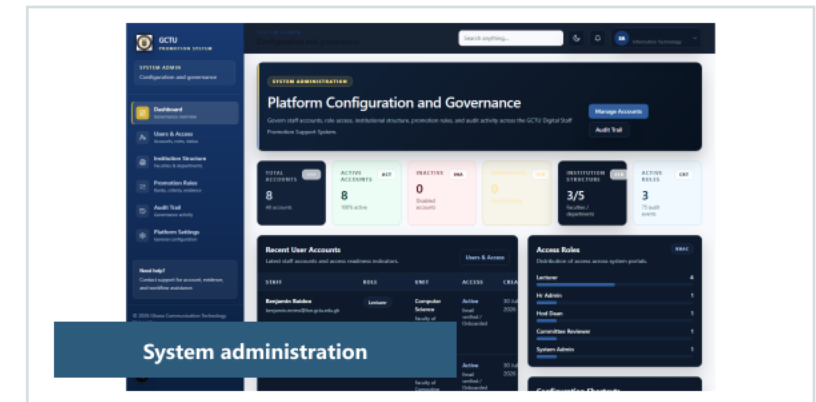
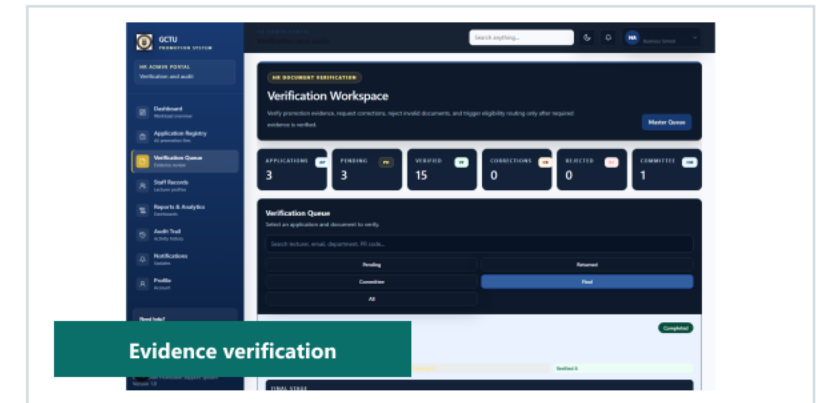
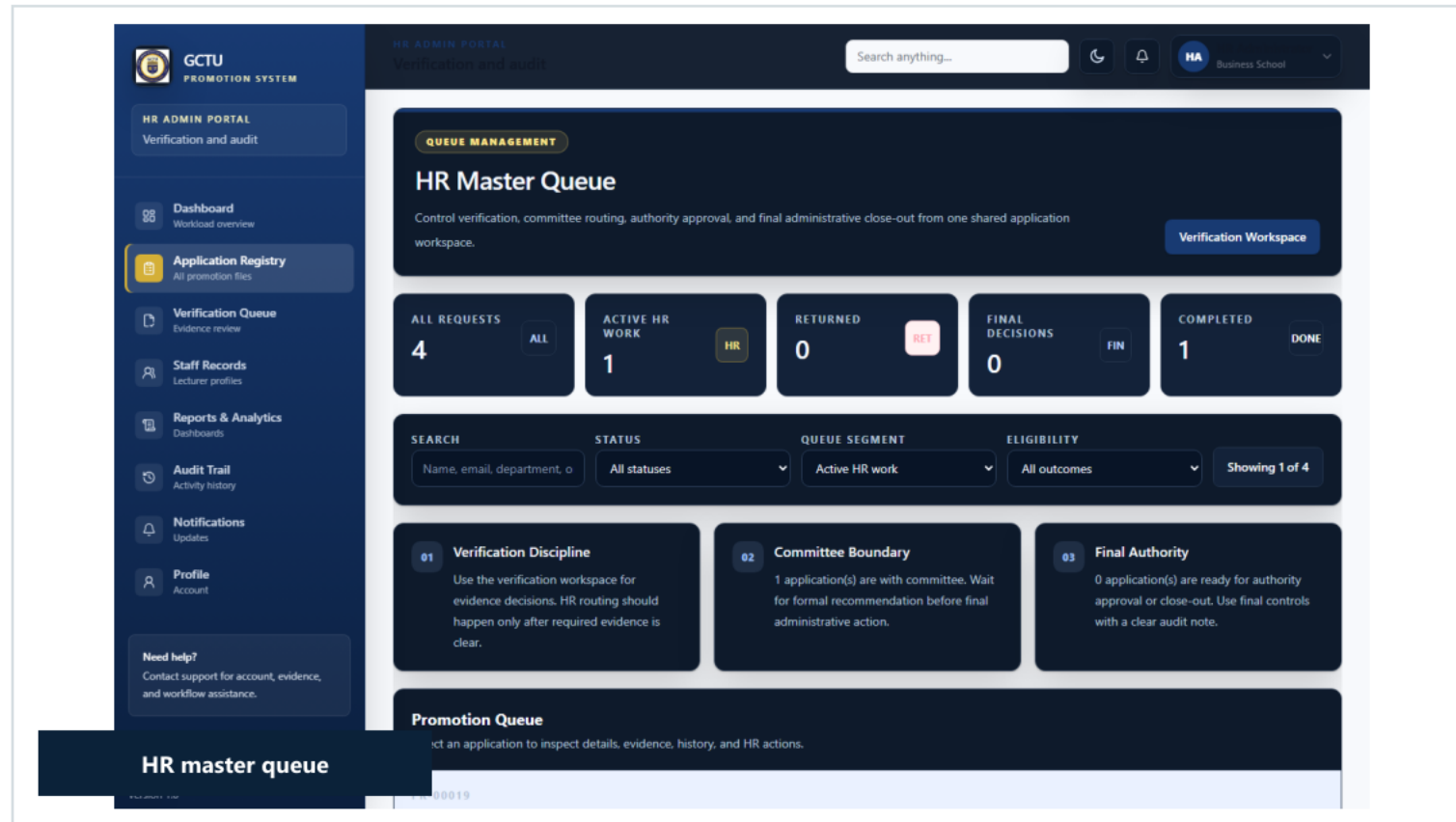
**Is the evidence academically strong,
relevant, and suitable for promotion?**

100/100 = verified core completeness

It is not a quality grade and not the final decision.

Implemented role-specific workspaces

Real interfaces from the completed prototype



Clear next actions, searchable queues, document preview, status labels, and organisational scope.

Testing changed the system, not just the report

36

formal functional and
integration tests

Type safety

TypeScript
verification

Build

Production
compilation

Database

Health and
seed checks

Browser

Complete role
workflow

Responsive

1440 px and
390 px views

DEFECT FOUND THROUGH END-TO-END TESTING

Initial result: verified evidence could still produce a zero score.

Root cause: the calculation depended on the wrong data source.

Correction: calculate directly from HR-verified document categories.

Result: workflow evidence directly improved implementation correctness.

Limitations define the production roadmap

CURRENT PROTOTYPE BOUNDARY

Completeness score only

Combined HOD / Dean stage

Academic pathway only

Criteria overwritten in place

UAT materials prepared

Standalone authentication

NEXT PRODUCTION STEP

Validate qualitative Schedule J assessment

Create sequential HOD then Dean routing

Add Schedule K staff pathway

Version rules by effective date

Run formal UAT with GCTU officers

Integrate SSO and staff records

Priority: institutional validation first, then statutory routing and criteria versioning.



CONCLUSION

A secure, centralised, and auditable promotion support workflow for GCTU

01

Standardises

evidence and workflow

02

Controls

role and organisational access

03

Preserves

human judgement and accountability

The system supports promotion decisions. It does not replace decision-makers.